MATH 111 Calculus 1

The precise definition of a limit

Week 1 · Lecture 3 · Stewart §1.7

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

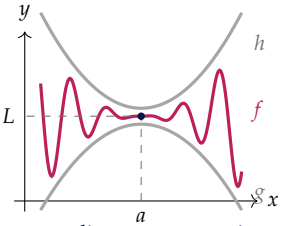
While you wait

Sit with your team. Notebook open: today you will write a proof together. Two volunteers as runners? Hands up.

From your exit tickets (1): what the Squeeze Theorem says — and does not say

Squeeze Theorem

If $g(x) \le f(x) \le h(x)$ for all x near a (except possibly at a), and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$, then $\lim_{x \to a} f(x) = L$.



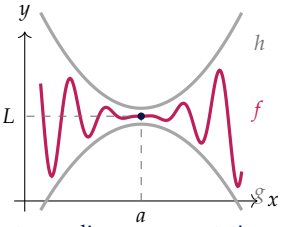
two policemen, one station

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Same station. The two policemen must have the *same* limit L; if the outer limits differ, the theorem says nothing at all.



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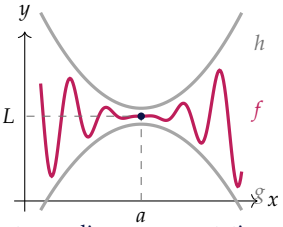
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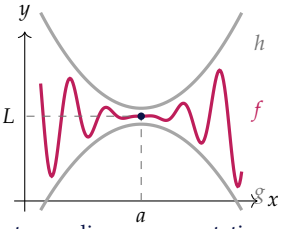


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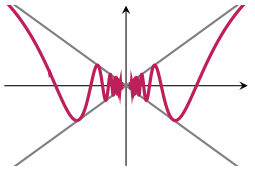
Near a, not at a. The inequality is needed only close to a, on both sides; $f(a)$ may be anything, or undefined.

You never compute $\lim f$. The theorem is for functions you *cannot* compute: you compute the two outer limits, and f has no choice.

From your exit tickets (2): how the theorem gives $\lim_{x\to 0} x \sin\frac{1}{x} = 0$

Step 1 — the wild part. $-1 \le \sin\frac{1}{x} \le 1$ for every $x \neq 0$. That is *all* we know — $\sin\frac{1}{x}$ itself has no limit at 0.

$y = x\sin\frac{1}{x}$ between $y = \pm x$



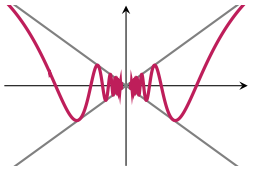
the amplitude $|x|$ shrinks to 0

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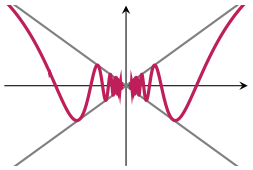
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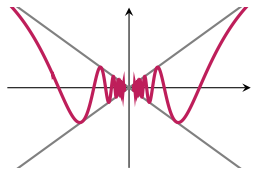
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Step 4 — conclude. By the Squeeze Theorem, $\lim_{x\to 0} x \sin\frac{1}{x} = 0$. Not by taming the oscillation — by shrinking its amplitude.

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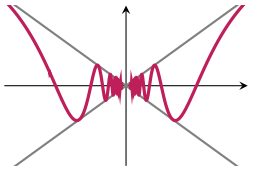
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Contrast. For $\sin\frac{1}{x}$ alone the policemen are -1 and 1 — different stations: the theorem gives nothing, and indeed there is no limit.

$y = x \sin\frac{1}{x}$ between $y = \pm x$



the amplitude $|x|$ shrinks to 0

STAND UP

"$\lim_{x \to a} f(x) = L$ means: $f(x)$ gets closer and closer to L as x gets closer and closer to a."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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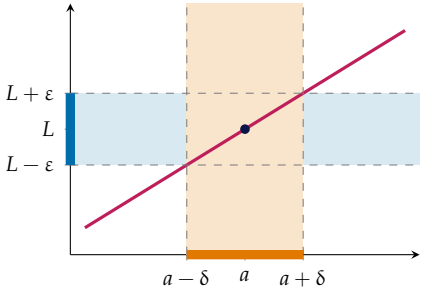
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Verdict

Not good enough. Take $f(x) = x^2$ near $a = 0$. As $x \to 0$, the distance $|x^2 - (-1)| = x^2 + 1$ *decreases*, so x^2 "gets closer and closer" to -1 too — yet the limit is certainly not -1. "Closer and closer" must become "as close as we like" — or, as Cauchy wrote in 1821, the values must "end by differing from it by as little as one wishes".

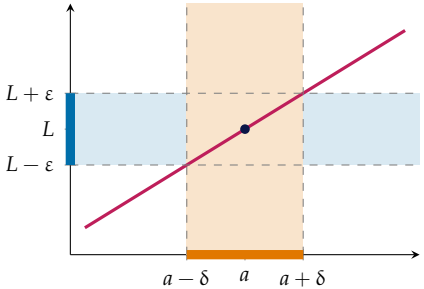
The ε–δ game



Whenever x is in the orange interval $(a-\delta, a+\delta)$ —
and $x \neq a$ — $f(x)$ is in the blue interval $(L-\varepsilon, L+\varepsilon)$.

$$f(x) = 2x - 1,\ a = 3,\ L = 5.$$

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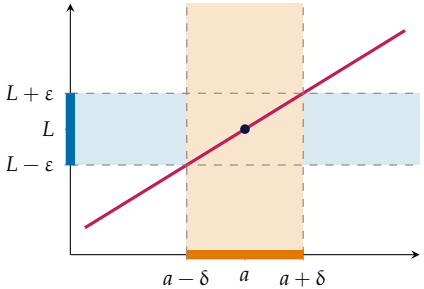


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You challenge: "get $f(x)$ within ε of 5."

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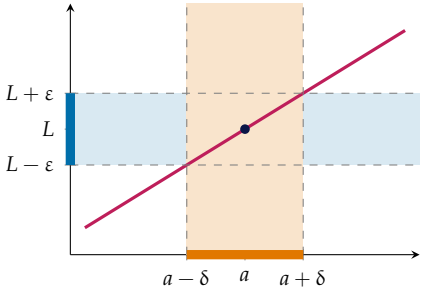
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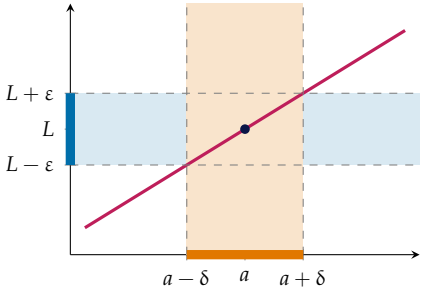
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my δ	0.05	0.005	0.0005	$\varepsilon/2$

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Why $\varepsilon/2$? $|f(x) - 5| = |2x - 6| = 2|x - 3| < \varepsilon \iff |x - 3| < \varepsilon/2$.

A little history · 1823

The letters are Cauchy's. In his 1823 lectures he wrote the first proofs with an ε and a δ; the usual explanation is that ε stands for *erreur* — the error you are willing to tolerate — and δ for *différence*, the distance you are allowed to move. Two hundred years later every calculus student in the world still plays the game with the same two Greek letters.

Question 6

$f(x) = 2x - 1$, $a = 3$, $L = 5$. The challenge is $\varepsilon = 0.01$. Which δ works, i.e. guarantees $|f(x) - 5| < 0.01$ whenever $0 < |x - 3| < \delta$?

A. $\delta = 0.02$ **B.** $\delta = 0.01$ **C.** $\delta = 0.005$ **D.** $\delta = 1$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

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120 sec

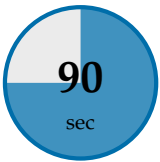
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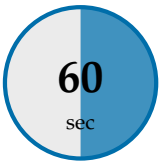
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Hands down. Thank you.



MIC

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the team nearest the left aisle in row 6 — runner, please. Give the reason; I repeat what you say.

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Answer

C. $|f(x) - 5| = 2|x - 3|$, so we need $|x - 3| < 0.005$. With $\delta = 0.01$ (B) the point $x = 3.008$ has $|f(x) - 5| = 0.016 > 0.01$. Any $\delta \leq 0.005$ works — smaller is always safe, larger is not.

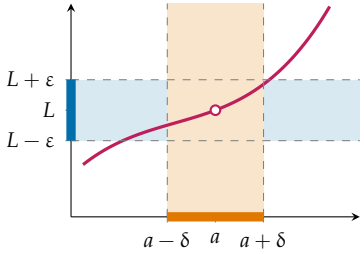
The precise definition

Definition

Let f be defined on an open interval containing a, except possibly at a. We say $\lim_{x \to a} f(x) = L$ if

for every $\varepsilon > 0$ there is a $\delta > 0$ such that

if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.



$|x - a| < \delta$ on the x-axis, $|f(x) - L| < \varepsilon$ on the y-axis; the value at a plays no role.

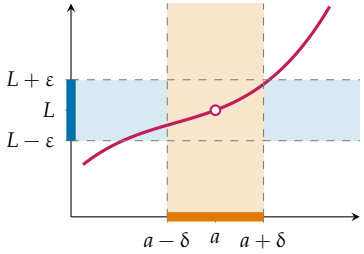
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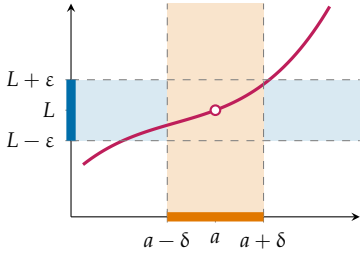
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- **Read it aloud together, twice:** "for every epsilon there is a delta …"
- $0 < |x - a|$ excludes $x = a$: the value $f(a)$ plays no role.



$|x - a| < \delta$ on the x-axis, $|f(x) - L| < \varepsilon$ on the y-axis; the value at a plays no role.

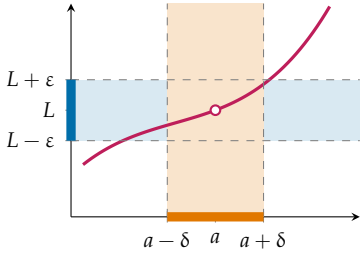
Definition

Let f be defined on an open interval containing a, except possibly at a. We say $\lim_{x \to a} f(x) = L$ if

for every $\varepsilon > 0$ there is a $\delta > 0$ such that

if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

- **Read it aloud together, twice:** "for every epsilon there is a delta …"
- $0 < |x - a|$ excludes $x = a$: the value $f(a)$ plays no role.
- δ is chosen *after* ε and may depend on it; "for every" and "there is" cannot be swapped. In the picture: for every blue interval around L there is an orange interval around a whose graph lies inside the band.



$|x - a| < \delta$ on the x-axis, $|f(x) - L| < \varepsilon$ on the y-axis; the value at a plays no role.

A little history · 1684–1861

Newton and Leibniz calculated with "infinitely small" quantities and got the right answers without being able to say what those quantities were. In 1734 Bishop Berkeley mocked them as "ghosts of departed quantities". Cauchy put the definition into words in 1821; Weierstrass turned it into ε and δ in his Berlin lectures around 1861. Calculus worked for 180 years before anyone could state precisely what a limit is — if today feels hard, you are in excellent company.

How to write an ε–δ proof

Claim. $\lim_{x \to 1}(6x + 2) = 8$.

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1. Scratch work — find δ

We want $|f(x) - 8| < \varepsilon$.

$$|f(x) - 8| = |(6x + 2) - 8|$$
$$= |6x - 6| = 6|x - 1|.$$

So $6|x - 1| < \varepsilon \iff |x - 1| < \varepsilon/6$.

Choose $\delta = \varepsilon/6$.

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2. The proof — write it forwards

Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon/6$.

If $0 < |x - 1| < \delta$, then

$$|(6x + 2) - 8| = 6|x - 1| < 6\delta = \varepsilon.$$

Hence $\lim_{x \to 1}(6x + 2) = 8$. $\square$

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Hence $\lim_{x \to 1}(6x + 2) = 8$. $\square$

The scratch work goes backwards (from what we want to what δ must be); the proof goes forwards (from δ to the conclusion). Both parts belong on your paper.

Question 7

Which statement is the correct meaning of $\lim_{x \to a} f(x) = L$?

A. For every $\delta > 0$ there is $\varepsilon > 0$: if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

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In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 2 minutes

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Which is the correct meaning of $\lim_{x \to a} f(x) = L$? (same four statements, in symbols)

A. $\forall \delta > 0 \; \exists \varepsilon > 0 : \; 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

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A B C **D**

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Hands down. Thank you.

A B C D

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The microphone goes to

the team in the right block, row 3 — runner, please. Give the reason; I repeat what you say.

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Answer

B. A swaps the roles (ε must be the challenge); it is satisfied by *every* f bounded near a. C says one δ serves all ε, which forces f to be constant near a. D includes $x = a$, so it would require $f(a) = L$ — that is continuity, not the limit.

Your turn: a proof in your notebook

TEAM WORK · 5 minutes

Prove from the definition: $\lim_{x \to 2}(3x + 1) = 7$.

In your team

Scribe writes; everyone dictates. Two parts: scratch work (find δ), then the proof written forwards. Sceptic checks every inequality.



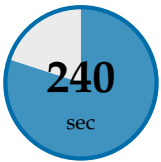
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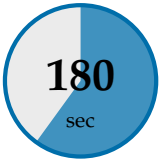
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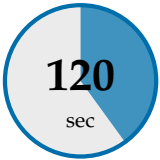


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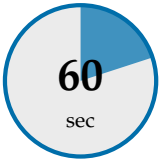
Your turn: a proof in your notebook

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Prove from the definition: $\lim_{x \to 2}(3x + 1) = 7$.

In your team

Scribe writes; everyone dictates. Two parts: scratch work (find δ), then the proof written forwards. Sceptic checks every inequality.



TEAM WORK · 5 minutes

Prove from the definition: $\lim_{x \to 2}(3x + 1) = 7$.

In your team

Scribe writes; everyone dictates. Two parts: scratch work (find δ), then the proof written forwards. Sceptic checks every inequality.



Your turn: a proof in your notebook

MIC

Prove from the definition: $\lim_{x \to 2}(3x + 1) = 7$. The courier reads the team's proof aloud, sentence by sentence.

The microphone goes to

the team in the middle block, row 10 — runner, please. Give the reason; I repeat what you say.

Prove from the definition: $\lim_{x \to 2}(3x + 1) = 7$. The courier reads the team's proof aloud, sentence by sentence.

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Answer

Model answer.

Scratch: we want $|f(x) - 7| < \varepsilon$; $|(3x + 1) - 7| = |3x - 6| = 3|x - 2| < \varepsilon \iff |x - 2| < \varepsilon/3$.

Proof: Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon/3$.

If $0 < |x - 2| < \delta$, then $|(3x + 1) - 7| = 3|x - 2| < 3\delta = \varepsilon$. Hence $\lim_{x \to 2}(3x + 1) = 7$. $\square$

Four things to tick in yours: "let $\varepsilon > 0$" first; a δ that depends on ε; the chain of (in)equalities; a concluding sentence.

A non-linear example

Claim. $\lim_{x \to 3} x^2 = 9$.

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Scratch work — find δ

We want $|x^2 - 9| < \varepsilon$.

$|x^2 - 9| = |x - 3|\,|x + 3|$. The factor $|x + 3|$ is not constant — tame it: if $|x - 3| < 1$, then $2 < x < 4$, so $|x + 3| < 7$.

Then $|x^2 - 9| < 7\,|x - 3|$, which is $< \varepsilon$ if $|x - 3| < \varepsilon/7$.

Two conditions on $|x - 3|$: take $\delta = \min(1, \varepsilon/7)$.

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Proof — write it forwards

Let $\varepsilon > 0$ be given. Choose $\delta = \min(1, \varepsilon/7)$.

If $0 < |x - 3| < \delta$, then $|x - 3| < 1$, so $|x + 3| < 7$, and

$$|x^2 - 9| = |x - 3|\,|x + 3| < 7\,|x - 3| < 7 \cdot \tfrac{\varepsilon}{7} = \varepsilon. \qquad \square$$

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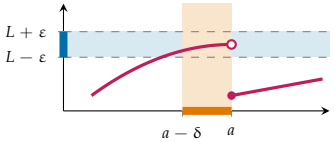
Moral: for non-linear f, first restrict δ (here $\delta \leq 1$) to tame the extra factor; then take min.

Precise versions of the other limits

One-sided limits

$\lim_{x \to a^-} f(x) = L$: for every $\varepsilon > 0$ there is $\delta > 0$ such that if $a - \delta < x < a$ then $|f(x) - L| < \varepsilon$.

$\lim_{x \to a^+} f(x) = L$: … if $a < x < a + \delta$ then $|f(x) - L| < \varepsilon$.



Left limit: only the half-strip $a - \delta < x < a$ counts.

Precise versions of the other limits

One-sided limits

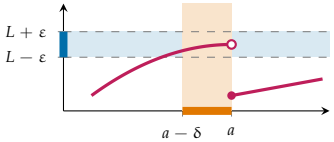
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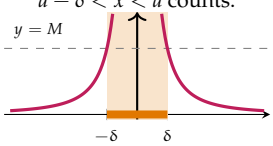
Infinite limits

$\lim_{x \to a} f(x) = \infty$: for every $M > 0$ there is $\delta > 0$ such that if $0 < |x - a| < \delta$ then $f(x) > M$.

(For $-\infty$: for every $N < 0$ … then $f(x) < N$.)



Left limit: only the half-strip $a - \delta < x < a$ counts.



$y = 1/x^2$: over the strip, the graph is above M.

Precise versions of the other limits

One-sided limits

$\lim_{x\to a^-} f(x) = L$: for every $\varepsilon > 0$ there is $\delta > 0$ such that if $a - \delta < x < a$ then $|f(x) - L| < \varepsilon$.

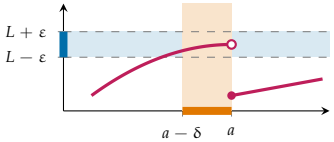
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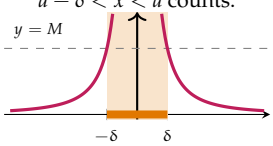
$\lim_{x\to a} f(x) = \infty$: for every $M > 0$ there is $\delta > 0$ such that if $0 < |x - a| < \delta$ then $f(x) > M$.

(For $-\infty$: for every $N < 0$ … then $f(x) < N$.)

Example. $\lim_{x\to 0} 1/x^2 = \infty$: given $M > 0$, $1/x^2 > M \iff |x| < 1/\sqrt{M}$. So $\delta = 1/\sqrt{M}$ works. □



Left limit: only the half-strip $a - \delta < x < a$ counts.



$y = 1/x^2$: over the strip, the graph is above M.

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(1) “In the definition, δ is allowed to depend on ε.”

(2) “If some δ works for a given ε, then every *larger* δ works too.”

Stand up if you think this is TRUE.

Stay standing until I say “sit”. Then find someone with the other opinion — 60 seconds.

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Verdict

(1) **TRUE** — that is the whole point; $\delta = \varepsilon/4$ above. (2) **FALSE** — every *smaller* δ works; a larger one lets in more x's and may break the inequality (Question 6: $\delta = 0.01$ failed where 0.005 worked).

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This week's quiz question comes from Questions 1–7

Homework: Stewart §1.6 Exercises 11–32 and 39–46; §1.7 Exercises 19–26 (linear ε–δ proofs). Tutorial this week: revision of functions (§1.1–1.3). Next week: continuity.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name (and number, once you have one); names of those present today.
2. One sentence: *"In the definition, ε controls … and δ controls …"*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you (or made you smile).

Pass it to the aisle; runners collect. Thank you.